Probability and Statistics 1
Chapter 1: Basic statistical notions and vocabulary

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1 Introduction

As explained in the course introduction, a statistical study follows a fixed chain: collection of raw observations, synthesis in a statistical table, visualisation, and numerical summarisation. This chapter is devoted to the first three stages of that chain, for a single character observed on a sample. We introduce the vocabulary common to every statistical study, classify characters into their different types, and then treat the frequency distribution of a qualitative character and of a quantitative character, together with the graphical representations attached to each case. The numerical summarisation of these distributions, through position and dispersion parameters, is the object of Chapter 2.

The vocabulary introduced here is not decorative. The distinction between a population and a sample, between a character and a modality, between a qualitative and a quantitative character, and between an absolute and a relative frequency, determines what may and may not be done with the data; every error of interpretation committed later in the course can be traced back to one of these distinctions being ignored.

2 The population and its elements

A statistical study is always relative to a well-delimited collection of objects: the students of a school, the servers of a datacenter, the repositories of an organisation, the requests received by a web service. This collection is called the population. Each of its members is an individual, and the part of the population that is actually observed is the sample. The three notions are defined below.

Definition.

The **population** is the complete set of units concerned by a statistical study. It is denoted by Ω.

Example.

1. To study the programming habits of the students of an artificial intelligence school, the population is the set of all students enrolled in that school.
2. To analyse the attacks received by a company's infrastructure, the population is the set of all connection attempts recorded by the intrusion-detection system during the observation period.

Definition.

An **individual**, or statistical unit, is one member of the population. We write $\omega \in \Omega$.

Example.

1. In the first population above, each student is an individual; if the students are labelled $s_1, \dots, s_N$, then $s_1 \in \Omega$.

2. In the second population, each connection attempt is an individual, even though it is not a human being: in statistics, an individual is whatever the study is about.

Definition.

A **sample** is a subset of the population on which the observations are actually made.

Example.

1. Instead of questioning all 1200 students of the school, we send the questionnaire to 80 of them, drawn at random; these 80 students form a sample of size $n = 80$.

2. Instead of labelling all 4×10^6 connection attempts of the year, the security team analyses a sample of 200 reported incidents.

Remark.

A population is not necessarily finite, nor even physical. The set of all requests that a server *could* receive, or the set of all possible outputs of a random-number generator, are perfectly legitimate populations. This point will be essential in Chapter 3, where Ω will denote the set of all possible outcomes of a random experiment, exactly as here it denotes the set of all units of the study.

Once the population is fixed, two strategies are possible: observe all of it, or observe only a part of it. The corresponding operations are the survey, the census and the poll.

Definition.

A **survey** is the operation consisting in observing, measuring or questioning the individuals of a sample, in order to collect data.

Example.

Distributing a questionnaire on code editors to the 80 students of the sample and recording the answers is a survey.

Definition.

A **census** is a survey in which the observed sample is the entire population (exhaustive survey).

Example.

Indexing every public repository of a code-hosting platform, or counting every inhabitant of a country, are censuses: the sample coincides with the population.

Definition.

A **poll** is a survey in which the observed sample is a strict subset of the population (non-exhaustive survey).

Example.

Asking 1000 randomly selected users what they think of a new feature of an application, out of several million users, is a poll.

Remark.

A census is exact but expensive, sometimes impossible (the population may be infinite, or destroyed by the observation itself). A poll is cheaper but introduces a *sampling error*, whose quantification is the object of inferential statistics. The choice between the two is a trade-off between cost and precision, not a matter of taste.

3 The statistical character

On each individual of the sample we observe one or several characteristics. The characteristic under study is called the character; the values it takes are its modalities. The nature of the character dictates the whole subsequent treatment: the shape of the statistical table, the admissible graphs, and the parameters that may be computed.

Definition.

A **character** is a characteristic defined on the population and observed on the sample.

Example.

1. On a student, we may observe the character "main code editor" and the character "number of programming languages known".
2. On a reported security incident, we may observe the character "attack type" and the character "time to detection, in hours".

Definition.

The different values taken by a character are called **modalities**, or observations.

Example.

1. The modalities of the character "main code editor" are in our study e.g. VS Code, Jupyter Notebook, Vim/Neovim, emacs.
2. The modalities of the character "attack type" are Phishing, Ransomware, Credential stuffing, DDoS and Other.

Two large families of characters must be distinguished.

Definition.

A **qualitative character** is a character whose modalities are not measurable. It is said to be *nominal* when no natural order exists between its modalities, and *ordinal* when such an order

exists.

Example.

1. *Nominal:* main code editor (VS Code, Jupyter, Vim/Neovim, emacs), programming language (Python, C++, Rust, JavaScript), cloud provider (AWS, Azure, GCP), attack type (phishing, ransomware, DDoS). No modality is "greater" than another.
2. *Ordinal:* severity of a vulnerability (low, medium, high, critical), user satisfaction (1 to 5 stars). Here the modalities are ordered, but the differences between two consecutive modalities have no numerical meaning.

Definition.

A **quantitative character** is a character whose modalities are numbers, and which can therefore be measured. It is said to be *discrete* when it takes isolated values, generally integers, and *continuous* when it takes any value in an interval.

Example.

1. *Discrete:* number of GPUs in a cluster node, number of commits per day, number of failed login attempts, number of parameters of a model expressed in billions.
2. *Continuous:* inference latency of a model in milliseconds, size of a dataset in gigabytes, energy consumption of a datacenter in megawatt-hours, detection time of an intrusion in hours.

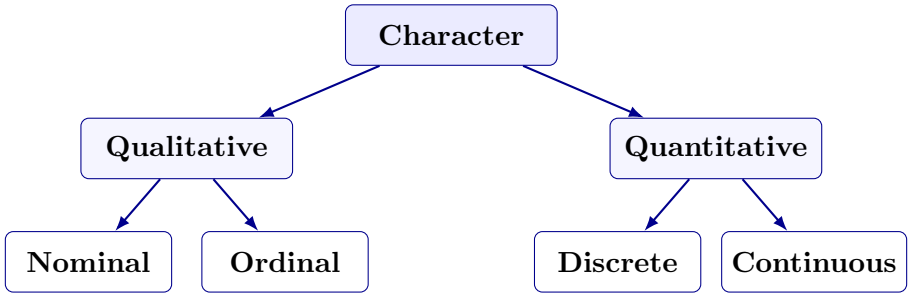


Figure 3.1: Typology of statistical characters.

Remark.

The fact that a modality is written with digits does not make the character quantitative. A student number, an HTTP status code or a hexadecimal colour code are labels: averaging them has no meaning. Symmetrically, a quantitative character may be recorded in classes and thus handled as an ordered qualitative character, as will be seen in Section 6.

In the remainder of this chapter we study a single character, observed on a sample of size n, whose k modalities $x_1, \dots, x_k$ form a *complete system*: every individual of the sample belongs to exactly one modality. We treat first the case of a qualitative character (Sections 4 and 5), then the case of a quantitative character (Sections 6 and 7). The counting of modalities and the construction of the table are common to the two cases; the graphs and the treatment of large numbers of distinct values differ.

4 Frequency distribution of a qualitative character

Raw observations are unusable as they stand: a list of 80 answers is not a summary. The first operation of descriptive statistics is therefore to *count* how often each modality appears, which leads to the three fundamental quantities below.

Definition.

The **absolute frequency** (or effectif) of the modality x_i, denoted n_i, is the number of individuals of the sample whose character takes the value x_i.

Example.

In the survey on code editors, 38 of the 80 students answered "VS Code"; the absolute frequency of this modality is therefore $n_{\text{VSC}} = 38$.

Definition.

The **relative frequency** of the modality x_i is the number

$$f_i = \frac{n_i}{n},$$

where n is the size of the sample.

Example.

For the same survey, $f_{\text{VSC}} = 38/80 = 0.475$.

Definition.

The **percentage** of the modality x_i is

$$p_i = \frac{n_i}{n} \times 100 = 100\, f_i.$$

Example.

$p_{\text{VSC}} = 0.475 \times 100 = 47.5$, which is read as $47.5\,\%$.

These three quantities are not independent, and they satisfy two properties that are used constantly as verification tools.

Proposition.

Let $n_1, \dots, n_k$ be the absolute frequencies of the k modalities of a qualitative character observed on a sample of size n. Then

$$\sum_{i=1}^{k} n_i = n, \qquad 0 \le f_i \le 1, \qquad \sum_{i=1}^{k} f_i = 1, \qquad \sum_{i=1}^{k} p_i = 100.$$

Indeed, each individual of the sample belongs to exactly one modality (the system is complete), so the counts add up to the size of the sample; dividing by n gives the second and third relations.

Remark. *Rounding*

The relations above hold for the exact values. If the frequencies are rounded, the sum may differ slightly from 1 or from 100. One then adjusts the largest frequency so that the total is exactly 1, rather than rounding each value independently.

The frequency distribution is presented in a table, whose construction is purely mechanical once the modalities have been fixed.

Definition.

A **frequency table** of a qualitative character is a table with one row per modality, giving the absolute frequency n_i, the relative frequency f_i and the percentage p_i of each modality, the last row containing the totals.

Example.

1. *Survey on code editors* ($n = 80$ first-year students of an AI school). The construction follows four steps: (i) list the modalities, including a residual modality "Other" if necessary; (ii) count each modality; (iii) divide by n; (iv) multiply by 100 and check the totals.

Main editor	n_i	$f_i = n_i/n$	$p_i = 100\, f_i$
VS Code	38	0.4750	47.50
Jupyter Notebook	18	0.2250	22.50
Vim/Neovim	14	0.1750	17.50
emacs	10	0.1250	12.50
Total	80	1.0000	100.00

Table 4.1: Frequency distribution of the main code editor ($n = 80$).

5 Graphical representations of a qualitative character

A table is precise but slow to read. Graphs make the shape of the distribution visible at a glance: which modality dominates, whether the distribution is concentrated or spread, whether a residual category is negligible. Two representations are standard for qualitative characters.

5.1 Pie chart (circular diagram)

Definition.

A **pie chart** represents the distribution on a disk divided into sectors whose areas, hence whose angles at the centre, are proportional to the frequencies. The angle of the sector of modality x_i is

$$\theta_i = \frac{n_i}{n} \times 360^\circ = \frac{p_i}{100} \times 360^\circ.$$

The formula follows from a proportionality argument: the full disk corresponds to n individuals and to 360°, hence one individual corresponds to $360^\circ/n$ and n_i individuals to $n_i \times 360^\circ/n$.

Proposition.

The angles defined above satisfy $\sum_{i=1}^{k} \theta_i = 360°$, and θ_i does not depend on the units used to express the frequencies.

Example.

1. *Code editors* ($n = 80$). The angles are

$$\theta_{\text{VSC}} = 0.4750 \times 360° = 171°, \quad \theta_{\text{Jup}} = 0.2250 \times 360° = 81°,$$

$$\theta_{\text{Vim}} = 0.1750 \times 360° = 63°, \quad \theta_{\text{emacs}} = 0.1250 \times 360° = 45°,$$

and their sum is $171 + 81 + 63 + 45 = 360$, as expected. The resulting disk is Figure 5.1.

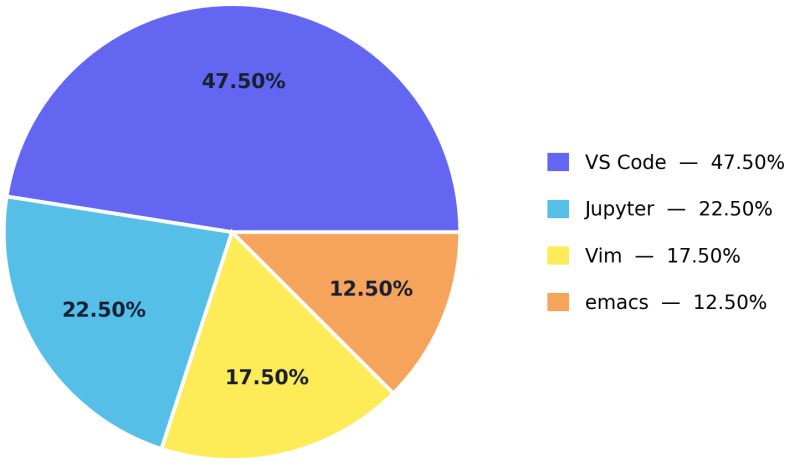


Figure 5.1: Pie chart of the main code editor ($n = 80$).

Remark.

A pie chart is meaningful only if the modalities form a partition of a whole, that is, if they are exhaustive and mutually exclusive. It becomes unreadable beyond about six sectors in general, and it does not allow a precise comparison of two close frequencies, whereas a bar chart does. It is therefore a communication tool rather than an analysis tool.

5.2 Bar chart

Definition.

A **bar chart** represents each modality by a vertical rectangle whose height is proportional to its absolute frequency n_i (or to f_i, or to p_i), the modalities being placed on the horizontal axis and the vertical axis starting at zero.

Example.

The frequency distribution of Table 4.1 is represented in Figure 5.2. The heights are $38, 18, 14, 10$ and the axis starts at 0, so that the ratio of two heights is equal to the ratio of the corresponding frequencies.

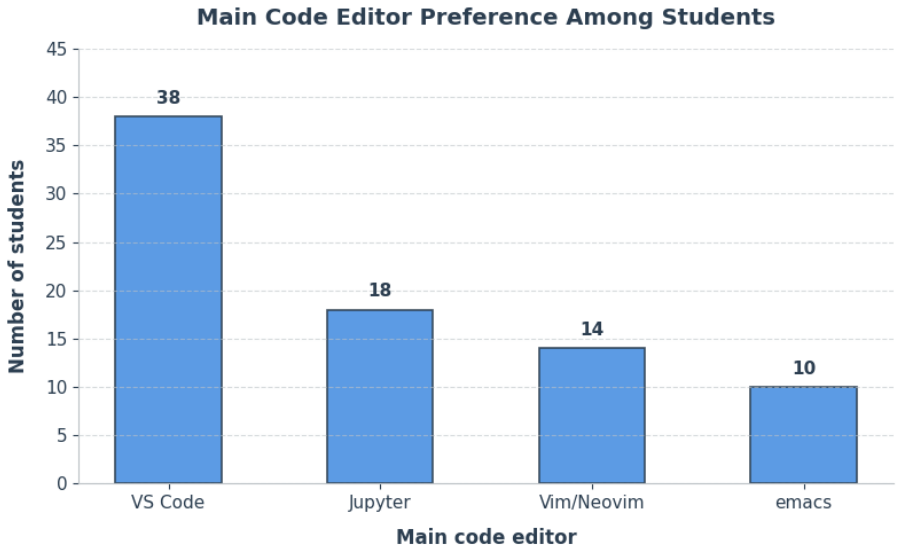


Figure 5.2: Bar chart of the main code editor ($n = 80$).

Remark.

The vertical axis must start at zero. A truncated axis exaggerates differences and is a classic manipulation of opinion graphics. By contrast, the *order* of the modalities is free: it is often useful to sort the bars by decreasing frequency, which gives a Pareto chart and immediately exhibits the dominant modalities.

6 Frequency distribution of a quantitative character

The counting operations are the same as in the qualitative case: an absolute frequency n_i, a relative frequency $f_i = n_i/n$, a percentage $p_i = 100\, f_i$, a total equal to n. What changes is the set of values that the character may take, and consequently the shape of the table.

For a *discrete* quantitative character, the number of distinct values is finite and generally small: the number of GPUs per node takes the values $1, 2, 4, 8, 16$, the number of commits per day takes integer values. The table is then built exactly as for a qualitative character, with the modalities replaced by the numerical values, sorted in increasing order.

For a *continuous* quantitative character, the situation is different. The number of possible values is in principle infinite, and the observations are rarely repeated, because the precision of the measurement limits the number of distinct values that can be distinguished. A latency of 42.37 ms and a latency of 42.38 ms are distinct observations, and a table with one row per observed value would have as many rows as the sample has individuals. The data are therefore grouped into *classes*, that is, into intervals.

Definition.

A **class** is an interval of the form $[a_i, b_i[$ (or $]a_i, b_i]$) into which the values of a continuous quantitative character are grouped. Its **centre** is the number

$$x_i = \frac{a_i + b_i}{2},$$

and its **amplitude** (length or range) is the number $b_i - a_i$. The **absolute frequency** of the class is the number n_i of observations that fall inside it.

Example.

For the inference latency of a model, measured in milliseconds on $n = 100$ requests, we use the classes $[30; 40[$, $[40; 50[$, $[50; 60[$, $[60; 70[$, $[70; 80[$, all of amplitude 10. The centre of the third class is $(50 + 60)/2 = 55$ ms.

The number of classes K is chosen by the statistician, subject to two constraints: the classes must be disjoint, and their union must contain all the observations. Two rules of thumb are used in practice.

Proposition. *Choice of the number of classes*

For a sample of size n, a number of classes close to

$$K = \begin{cases} \sqrt{n} & \text{if } n < 50, \\ 1 + \dfrac{10}{3} \log_{10} n & \text{if } n \geq 50 \end{cases}$$

gives a readable table. It is recommended to choose an odd number, and to give all classes the same amplitude

$$a = \frac{r}{K}, \qquad r = x_{\max} - x_{\min},$$

where r is the range of the series.

The rule is not a theorem: it only prevents the two degenerate situations, a table with too many rows to be read and a table with too few rows to show the shape of the distribution. It is standard to round the amplitude a to a convenient value, and to let the first and last classes extend slightly beyond the observed minimum and maximum.

Definition.

The **increasing cumulative absolute frequency** of modality or class i is

$$n_i^{\nearrow} = n_1 + n_2 + \cdots + n_i,$$

the number of observations whose value is less than or equal to the upper bound of class i. The **decreasing cumulative absolute frequency** is defined symmetrically, starting from the last class.

Example.

For the latency data of Table 6.2, the increasing cumulative frequencies are $8, 30, 71, 92, 100$: 71 of the 100 requests were served in less than 60 ms, and all of them in less than 80 ms.

Example.

1. *Discrete quantitative character: number of GPUs per node* of a compute cluster, observed on $n = 60$ nodes.

GPUs x_i	n_i	f_i	p_i	$n_i^{\nearrow}$	$f_i^{\nearrow}$
1	12	0.200	20.0	12	0.200
2	20	0.333	33.3	32	0.533
4	16	0.267	26.7	48	0.800
8	9	0.150	15.0	57	0.950
16	3	0.050	5.0	60	1.000
Total	60	1.000	100.0	–	–

Table 6.1: Frequency distribution of the number of GPUs per node ($n = 60$).

2. *Continuous quantitative character: inference latency* of a model, measured in milliseconds on $n = 100$ requests. The range is $r = 80 - 30 = 50$, and the rule above gives $K = 1 + \frac{10}{3}\log_{10} 100 = 7.67$, rounded to $K = 5$ classes of amplitude $a = 10$.

Class	Centre x_i	n_i	f_i	$n_i^{\nearrow}$
$[30; 40[$	35	8	0.08	8
$[40; 50[$	45	22	0.22	30
$[50; 60[$	55	41	0.41	71
$[60; 70[$	65	21	0.21	92
$[70; 80[$	75	8	0.08	100
Total	–	100	1.00	–

Table 6.2: Frequency distribution of the inference latency, grouped in classes ($n = 100$).

Remark. *Discretisation*

The centre x_i represents the class $[a_i, b_i[$ in all the computations of Chapter 2. In physics one assimilates any object to its centre of gravity; here one assimilates any observation of the class to the centre of the class. This operation is called *discretisation* of the continuous case, and it allows the formulas established for a discrete character to be applied without change. It is also the source of a small approximation error, which is negligible when the classes are narrow and all of the same amplitude.

Remark. *Loss of information*

A table of classes does not reproduce the original series: two datasets with the same class counts may have different means and different variances. What is gained is legibility, and robustness against isolated extreme values, which are absorbed into the first or the last class. What is lost is the exact position of each observation within its class. The choice of classes is therefore an editorial decision as much as a statistical one, and it must be reported together with the table.

7 Graphical representations of a quantitative character

For a discrete quantitative character, the graphs of the qualitative case remain valid, with the modalities replaced by the numerical values, sorted in increasing order. For a continuous quantitative character grouped in classes, the histogram replaces the bar chart, and the cumulative curve replaces the cumulative bar chart.

7.1 Bar chart for a discrete variable

Definition.

A **bar chart** for a discrete quantitative variable represents each value x_i by a vertical segment, or a thin rectangle, whose height is proportional to n_i. Unlike the bar chart of a qualitative character, the values are placed on the axis in their natural numerical order, so that the spacing between the bars carries information.

Example.

The distribution of Table 6.1 is represented in Figure 7.1. The values $1, 2, 4, 8, 16$ are not equally spaced, and the spacing of the bars reflects this: the bar of value 16 is visibly far from its neighbours, which is a feature of the distribution and not an accident of drawing.

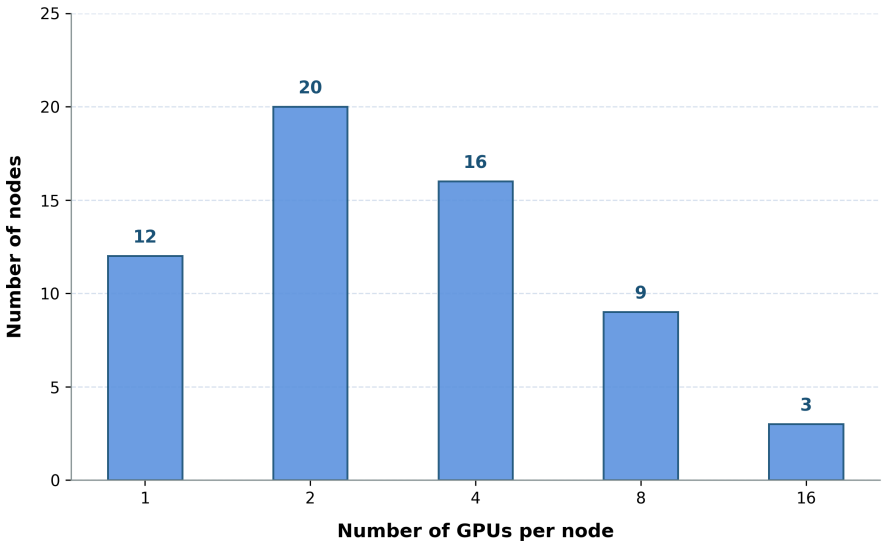


Figure 7.1: Bar chart of the number of GPUs per node ($n = 60$).

7.2 Polygon of frequencies

Definition.

The **polygon of frequencies** of a discrete quantitative variable is the broken line obtained by joining the points (x_i, n_i), completed on the left and on the right by the points $(x_1 - a, 0)$ and $(x_k + a, 0)$, where a is the common spacing of the values. It represents the same information as the bar chart, but as a profile.

Example.

The polygon of the GPU distribution, Figure 7.2, joins the points $(1, 12)$, $(2, 20)$, $(4, 16)$, $(8, 9)$, $(16, 3)$. Its shape shows immediately that the distribution is concentrated on small values, with a long tail towards the workstations equipped with many GPUs.

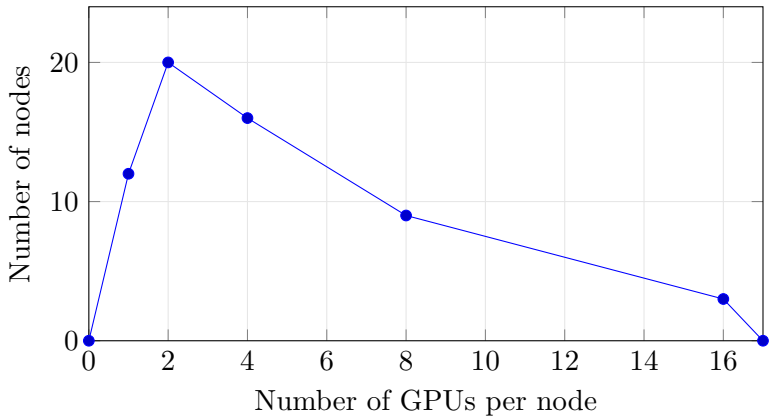


Figure 7.2: Polygon of frequencies of the number of GPUs per node ($n = 60$).

Remark.

The polygon is preferred to the bar chart when the variable is quantitative and the number of distinct values is large, because the broken line shows the profile of the distribution, its mode and its asymmetry, whereas a forest of bars hides them. For a qualitative character the polygon is meaningless: the order of the modalities is arbitrary, so the segments have no interpretation.

7.3 Histogram

Definition.

The **histogram** of a continuous quantitative character grouped in classes is the set of rectangles whose base is the class interval and whose height is the absolute frequency n_i of the class. When the classes have the same amplitude, the height may equivalently be n_i, f_i or p_i. When the amplitudes differ, the height must be proportional to n_i/a_i, so that the *area* of each rectangle remains proportional to the number of observations.

The rule on areas is not a technicality. It is what distinguishes a histogram from a bar chart: in a bar chart, the width of the rectangles is arbitrary and only the height carries meaning; in a histogram, the width is the amplitude of the class, and the area carries meaning. The total area of the histogram, divided by the common amplitude, is equal to the size of the sample.

Proposition.

If all the classes have the same amplitude a and the heights are the absolute frequencies n_i, the total area of the histogram is an, where n is the size of the sample. If the heights are the relative frequencies f_i, the total area is a.

Example.

The distribution of Table 6.2 is represented in Figure 7.3. All classes have amplitude 10 and the

heights are $8, 22, 41, 21, 8$; the total area is $10 \times (8 + 22 + 41 + 21 + 8) = 10 \times 100 = 1000 = an$, as stated. The modal class is clearly $[50; 60[$, with 41 of the 100 requests.

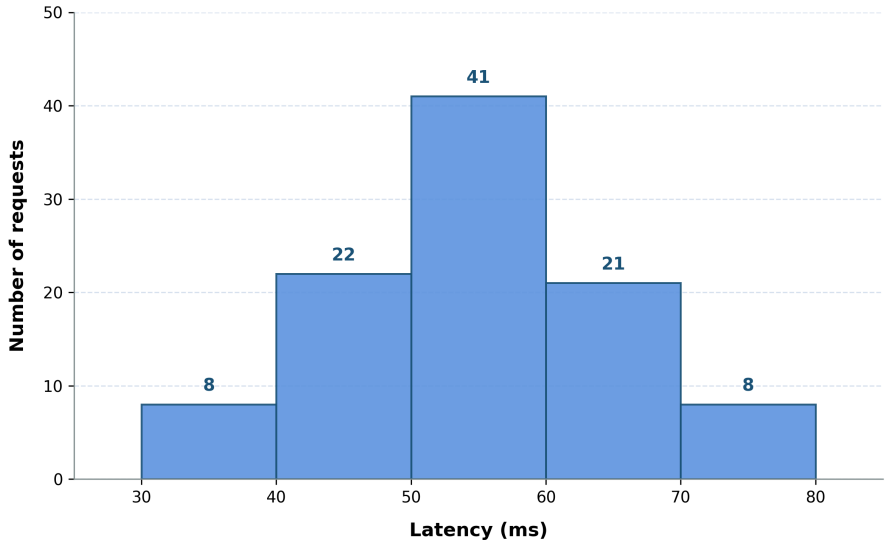


Figure 7.3: Histogram of the inference latency, classes $[30; 40[$ to $[70; 80[$ ($n = 100$).

Remark.

A histogram with fewer than five classes hides the shape of the distribution; with more than twenty classes, it shows the noise of the sample rather than the distribution. The rule of Proposition 6.2 is a guide, not an obligation, and a good histogram is often obtained by trying two or three numbers of classes and comparing the results.

7.4 Cumulative frequency curve

Definition.

The **cumulative frequency curve** of a continuous quantitative character is the graph of the increasing cumulative absolute frequency $n_i^{\nearrow}$, plotted against the upper bound of each class and joined by line segments. It is a non-decreasing curve, starting at 0 and ending at n.

Example.

For the latency data, the cumulative curve of Figure 7.4 passes through the points $(40, 8)$, $(50, 30)$, $(60, 71)$, $(70, 92)$, $(80, 100)$. It reads directly on the graph that $71\,\%$ of the requests are served in less than 60 ms, and that the median latency, the value for which half of the requests are served, lies in the class $[50; 60[$. This construction will be used systematically in Chapter 2 to compute the median and the quartiles of a continuous variable.

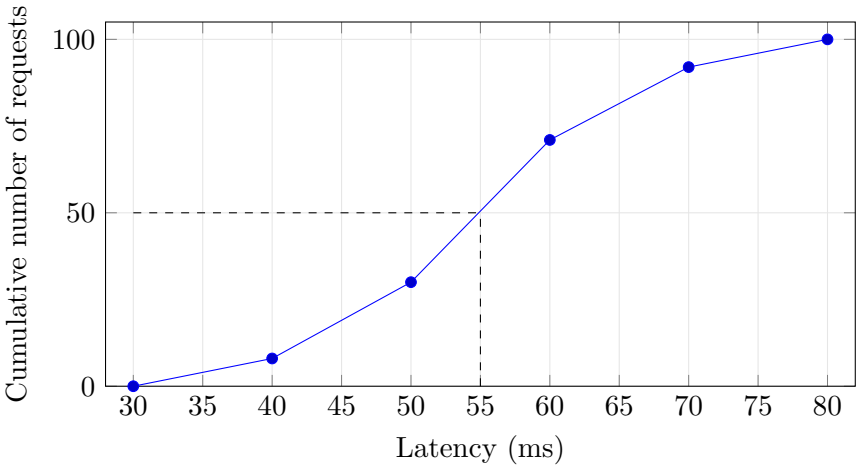


Figure 7.4: Cumulative frequency curve of the inference latency ($n = 100$).

Remark.

The cumulative curve answers questions of the form "how many observations are below this threshold?" and "below which threshold do half of the observations lie?". It is therefore the natural tool for the median and the quartiles, and it is the graphical counterpart of the cumulative frequencies of Table 6.2. It is also the object that will be turned into a distribution function in Chapter 3.

8 Common mistakes

The following errors appear every year in examinations and, more seriously, in professional reports.

1. **Forgetting the residual modality.** Without "Other", the frequencies do not sum to 1, and every graph drawn from the table is distorted.

2. **Truncating the axis of a bar chart.** The frequency axis must start at zero; otherwise the visual ratio of the bars no longer represents the ratio of the frequencies.

3. **Using a pie chart for non-exhaustive categories.** If the categories do not partition the population, the disk represents nothing and the angles are meaningless.

4. **Confusing bar chart and histogram.** In a bar chart, the width of the rectangles is arbitrary and only the height counts; in a histogram, the width is the amplitude of a class and the area counts. Drawing a histogram with unequal classes and equal heights is a mistake, not a stylistic choice.

5. **Treating codes as numbers.** Computing the mean of student numbers, of HTTP status codes or of hexadecimal colours is meaningless, because these characters are nominal. A digit is not a number.

6. **Choosing the classes after seeing the data.** The classes must be fixed before the table is built, and reported with it. Choosing them to make a particular class look dominant is a form of manipulation.

9 Summary

- A statistical study concerns a *population*, observes a *sample*, and is carried out through a *survey*, which is a *census* if it is exhaustive and a *poll* otherwise.
- A *character* is qualitative (nominal or ordinal) or quantitative (discrete or continuous); its values are its *modalities*. The type of the character determines the table, the graphs and the admissible parameters.
- For a character with k modalities observed on a sample of size n:

$$n_i, \qquad f_i = \frac{n_i}{n}, \qquad p_i = 100\, f_i, \qquad \sum_{i=1}^{k} n_i = n, \quad \sum_{i=1}^{k} f_i = 1, \quad \sum_{i=1}^{k} p_i = 100.$$

- For a *qualitative character*, the *frequency table* collects these quantities; the *bar chart*, the *band diagram* and the *pie chart* (with angles $\theta_i = (n_i/n) \times 360^\circ$) visualise them.
- For a *discrete quantitative character*, the same table applies, the modalities being replaced by numerical values sorted in increasing order; the *bar chart* and the *polygon of frequencies* replace the graphs of the qualitative case.
- For a *continuous quantitative character*, the data are grouped into *classes* of centre $x_i = (a_i + b_i)/2$ and amplitude $b_i - a_i$; the table adds the classes, their centres and the cumulative frequencies; the *histogram* and the *cumulative frequency curve* visualise them, the area of the histogram playing the role that the height plays in a bar chart.
- Chapter 2 attaches numbers to these representations: mode, mean, median, quartiles, variance, standard deviation and coefficient of variation, computed from the frequency tables built here.

References

[1] D. Freedman, R. Pisani, R. Purves, *Statistics*, 4th edition, W. W. Norton, 2007. Chapters 1–3 give a conceptual introduction to populations, samples, characters and frequency tables.

[2] D. C. Montgomery, G. C. Runger, *Applied Statistics and Probability for Engineers*, 7th edition, Wiley, 2018. Chapter 1 covers the vocabulary, the frequency tables and the graphs treated here, with engineering examples.

[3] J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th edition, Cengage, 2016. Chapter 1 gives the same material with a more formal treatment of the choice of classes.

[4] G. Saporta, *Probabilités, analyse des données et statistique*, 3rd edition, Technip, 2011. Chapters 1–2 are the French reference for the terminology used in these notes (effectif, classe, centre de classe, histogramme).

[5] J. W. Tukey, *Exploratory Data Analysis*, Addison-Wesley, 1977. The classical source for the idea that graphs are instruments of analysis and not merely illustrations.

[6] M. P. Deisenroth, A. A. Faisal, C. S. Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020. Chapter 6 shows how the histograms and cumulative curves of this chapter become empirical distributions in a machine-learning context.